MIL-C-18053A(Aer)
15 July 1958
Superseding
MIL-C-18053(Aer)
18 June 1954

MILITARY SPECIFICATION

CONTROL AMPLIFIER; ALTITUDE (P-1 AUTOMATIC PILOT/RADAR SET, AN/APN-22)

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 This specification covers certain design requirements and all performance requirements for altitude control amplifiers of one type only, for installation in the P-1 automatic pilot interconnection to radar set AN/APN-22.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publication of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Plating (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-C-5015	Connectors; Electrical "AN" Type
MIL-D-5028	Drawings and Data Lists; Preparation of Manufacturers' (For Production Aeronautical and Associated Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-I-6181	Interference Limits and Tests; Aircraft Electrical and Electronic Equipment
MIL-P-6837	Pilot, Automatic, P-1
MIL-S-6872	Soldering Process, General Specification for
MIL-L-6880	Lubrication of Aircraft; General Specification for
MIL-P-6906	Plates; Information and Identification
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery

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MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys

STANDARDS

Federal

FED-STD-1	Standard for Laboratory Atmospheric Conditions for Testing
FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of U.S. Military Property
MS33586	Metals, Definition of Dissimilar

DRAWINGS

Naval Aircraft Factory Standard Drawings

1347	Controller Altitude (P-1 Automatic Pilot/Radar Set AN/APN-22)
1348	Control Amplifier Altitude (P-1 Automatic Pilot/Radar Set AN/APN-22)

PUBLICATION

Air Force-Navy Aeronautical Bulletin

No. 143	Specifications and Standards; Use of

(When requesting any of the applicable documents refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, 700 Robbins Avenue, Philadelphia 11, Pennsylvania, Attention Code AD)

3. REQUIREMENTS

3.1 Preproduction - The control amplifier furnished under this specification shall be a product identically similar to one which has been tested and has passed the preproduction tests specified herein.

3.2 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.2.1 Critical materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

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3.2.2 Metals - Metals shall be of the corrosion-resistant type or shall be suitably protected as specified herein to resist corrosion in salt spray or atmospheric conditions to which the amplifier may be subjected when in storage or during normal service life.

3.2.2.1 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in Standard MS33586 shall not be used in intimate contact with each other.

3.2.3 Nonmagnetic materials - Nonmagnetic materials shall be used for all parts of the amplifier, except where magnetic materials are essential.

3.2.4 Fungus-proof materials - Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials which are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent, as approved by the Bureau of Aeronautics.

3.2.5 Selection of Government documents - Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.2.5.1 Standard parts - AN or MS parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used provided they have suitable properties and are replaceable by the AN or MS parts without alteration, and provided the corresponding AN or MS part numbers are referenced in the part lists and, if practicable, on the contractor's drawings. In applications for which no suitable corresponding AN or MS part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.3 Design and construction - The outline dimensions of the altitude control amplifier shall conform to NAF Drawing 1348, and shall be designed to operate with the P-1 type automatic pilot and Radar Set AN/APN-22. The altitude control amplifier shall consist of a suitable amplifier capable of amplifying signals received from the radar set and shall include a servo motor containing engaging clutches controlled by the amplifier output and shall also include a control operating from the servo system to supply signals to the elevator channel of the P-1 servo amplifier. The altitude control amplifier together with the height indicator ID-257/APN-22, shall be capable of maintaining an airplane by means of the automatic pilot within $\pm$10 feet of the preset altitude in the 0 to 500 foot range, and within 2 percent of the preset altitude in the 500 to 20,000 foot range of the radar set both in straight flight and in turns. These units shall also be capable of automatically controlling changes from one altitude to another throughout the range of the radar set, and shall include means to return the aircraft to level flight in the event of failure of the radar set, or the altitude control amplifier, or of the power supplied to either equipment during a climb or dive. The equipment shall also include a warning light to show such failure. The operation of the warning light to show failure of the equipment shall be limited to a power failure of the radar set or of the altitude control amplifier. The altitude control amplifier shall be so constructed as to withstand without failure the normal strain of jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service.

3.3.1 Anticipation control - Provision shall be made for an anticipation control to permit compensation of the device for the various time lag response characteristics that exist in different types of aircraft.

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3.3.2 Housing - The amplifier shall be provided with a housing which will allow adequate ventilation. The housing shall have removable covers for vacuum tube inspection, and shall have a lusterless black finish, Color No. 37038 of FED-STD-595.

3.3.3 Electrical receptacles - The electrical receptacles used shall conform to Specification MIL-C-5015 and shall be of the type specified on NAF Drawing 1347.

3.3.4 Mounting base - The control amplifier shall be provided with a mounting base conforming to NAF Drawing 1347. The mounting base shall include vibration insulators.

3.4 Performance - The amplifier shall perform satisfactorily when subjected to the following conditions and shall satisfy the performance requirements when subjected to the applicable tests as specified in section 4.

(a) Temperature - Temperature ranging from -54°C to 71°C.
(b) Altitude - Pressure ranging from 30 inches of mercury down to 7.04 inches of mercury. (35,000 feet altitude)
(c) Humidity - Relative humidity up to 100 percent.
(d) Fungus - Fungus growth as encountered in tropical climates.
(e) Salt spray - Exposure to salt sea atmosphere.
(f) Vibration - Vibrations incident to service use.

3.5 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.6 Special tools - The design of the control amplifier shall be such as to accommodate to the greatest possible extent, disassembly, reassembly, and service maintenance by those tools and items of maintenance equipment which are normally available as commercial standards.

3.7 Physical requirements

3.7.1 Weight - The weight of the amplifier shall not exceed 13 pounds.

3.7.2 Finish - Protective co ' ngs and finishes which will crack, chip or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.7.2.1 Aluminum alloy parts - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.7.2.2 Brass and steel parts - Where practicable, brass and steel parts shall be cadmium or zinc plated in accordance with Specification QQ-P-416 or QQ-Z-325.

3.7.3 Soldering - Soldering shall be performed in accordance with Specification MIL-S-6872.

3.7.4 Lubrication - Lubrication shall be accomplished in accordance with Specification MIL-L-6880 except that the selection of the lubricant used shall be approved by the Bureau of Aeronautics.

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3.7.5 Screw threads - Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL-S-7742.

3.8 Marking - All marking shall be durable to withstand usage encountered in service.

3.8.1 Identification of product - A nameplate conforming to Specification MIL-P-6906, shall be securely attached to the instrument case, and shall be legibly and permanently marked with the following information in accordance with Standard MIL-STD-130:

BUAER - U.S. NAVY
CONTROL AMPLIFIER; ALTITUDE
(P-1 AUTOMATIC PILOT/RADAR SET AN/APN-22)
NAF Part No. 1348-1
Federal Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract No. (or Order No. if Contract No. is not assigned)
Manufacturer's Name or Trademark

3.8.1.1 Manufacturer's part number - The manufacturer's part number marked in the space provided in the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.8.2 Receptacles - The electrical receptacle code markings as specified on NAF Drawing 1348 shall be legibly and permanently applied to the control amplifier housing.

3.9 Installation

3.9.1 Installation instructions - The contractor shall furnish with each altitude control amplifier one printed copy of instructions, with illustrations and diagrams if necessary, covering the installation of the amplifier. Prior to printing, two copies shall be furnished to the Bureau of Aeronautics for approval. Instructions shall be printed on 8-1/2 by 11 inches or 11 by 17 inch paper.

3.9.2 Envelope - An envelope furnished by the contractor containing the installation instructions shall be packaged with each instrument, and shall be marked with the following information:

"IMPORTANT
THIS ENVELOPE CONTAINS INSTRUCTIONS"

3.10 Workmanship - The amplifier including all parts and accessories, shall be constructed and finished to produce an instrument free from defects which affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests - The inspection and testing of the amplifiers shall be classified as follows:

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(a) Preproduction tests: Preproduction tests are those tests accomplished on samples which are representative of the production of the item after the award of contracts to determine that the production meets the requirements of this specification.

(b) Acceptance tests: Acceptance tests are those tests performed on amplifiers manufactured and submitted for acceptance under contract.

4.2 Preproduction tests - The preproduction tests of the amplifiers shall consist of all tests of this specification. The tests shall be conducted in the following sequence:

Individual tests
Sampling tests
Radio interference
Low temperature exposure
Altitude
Operation endurance
Vibration endurance
Humidity
Fungus resistance
Salt spray

4.2.1 Preproduction test samples - Preproduction test samples shall consist of three amplifiers of each manufacturer's part number submitted as samples for testing. After obtaining authorization for submittal from the Bureau of Aeronautics, the preproduction test samples shall be forwarded to the Supply Officer, Naval Air Development Center, Johnsville, Pennsylvania, Attention: Director, Aeronautical Instruments Laboratory. Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

Sample for preproduction test
Submitted by (name) (date) for preproduction tests in accordance with the requirements of Specification MIL-C- under authorization (reference letter authorizing test)
CONTROL AMPLIFIER; ALTITUDE (P-1 AUTOMATIC PILOT/RADAR SET, AN/APN-22)
Manufacturer's Part No.
Name of Manufacturer

4.3 Acceptance tests - The acceptance tests of the amplifiers shall consist of the individual tests and the sampling tests of this specification.

4.3.1 Individual tests - The individual tests of the amplifiers shall consist of the following tests conducted on each amplifier:

Inspection
Continuity
Output signal (sensitivity test)
Circuit reliability
10 cycle test

4.3.2 Sampling tests - The sampling tests of the amplifiers shall consist of the following tests conducted on each sampling test sample:

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Dielectric strength
Vibration error
Low temperature operation
High temperature operation

4.3.2.1 Sampling test instructions - Samples shall consist of three amplifiers selected at random by the inspector from each lot of 100 or less amplifiers which have passed the Individual tests. A lot shall consist of identical amplifiers with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at substantially the same time. Amplifiers which have been subjected to the sampling tests shall not be delivered on contract until they have been rebuilt and resubmitted to all Individual tests.

4.4 Rejection and retest - Items which have been rejected as a result of the Individual tests may be reworked to correct the defects, or replaced, and resubmitted for acceptance. Failure of samples representing lots shall be cause for rejection of the lot represented. Befor resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units or lots rejected after retest shall not be resubmitted without specific approval of the Bureau of Aeronautics.

4.5 Test conditions

4.5.1 Atmospheric conditions - Unless otherwise specified, all tests required by this specification shall be made in accordance with the requirements of FED-STD-1.

4.5.2 Vibration stand - Whenever a vibration stand is specified, it shall consist of a device which will vibrate at a desired frequency between 300 and 3000 cycles per minute and shall subject the amplifier to such vibration that a point on the vibration stand shall describe, in a plane inclined 45 degrees to the horizontal plane, a circle of the diameter specified herein.

4.5.3 Position - Unless otherwise specified, the control amplifier shall be tested in its normal operating position.

4.6 Test methods

4.6.1 Inspection - Each amplifier shall be examined to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.6.2 Continuity - The following circuits shall be tested for continuity. The resistance of each circuit shall be substantially zero.

From	To
J-2B	J-3E
J-2C	J-31
J-2E	Ground
J-3D	Ground
J-3G	Ground
J-5G	Ground

4.6.3 Output signal (sensitivity test) - The control amplifier shall be connected to an altitude controller, conforming to NAF Drawing 1347, as shown in Figure 1, with the "increase attitude" set at position 5 and one volt between Pin M of J-5 and ground. The output voltage between terminals A-D of J-2 shall be 260 $\pm$ 40 millivolts rms. The input voltage shall be reversed and the output voltage shall read essentially the same value. This test shall be made

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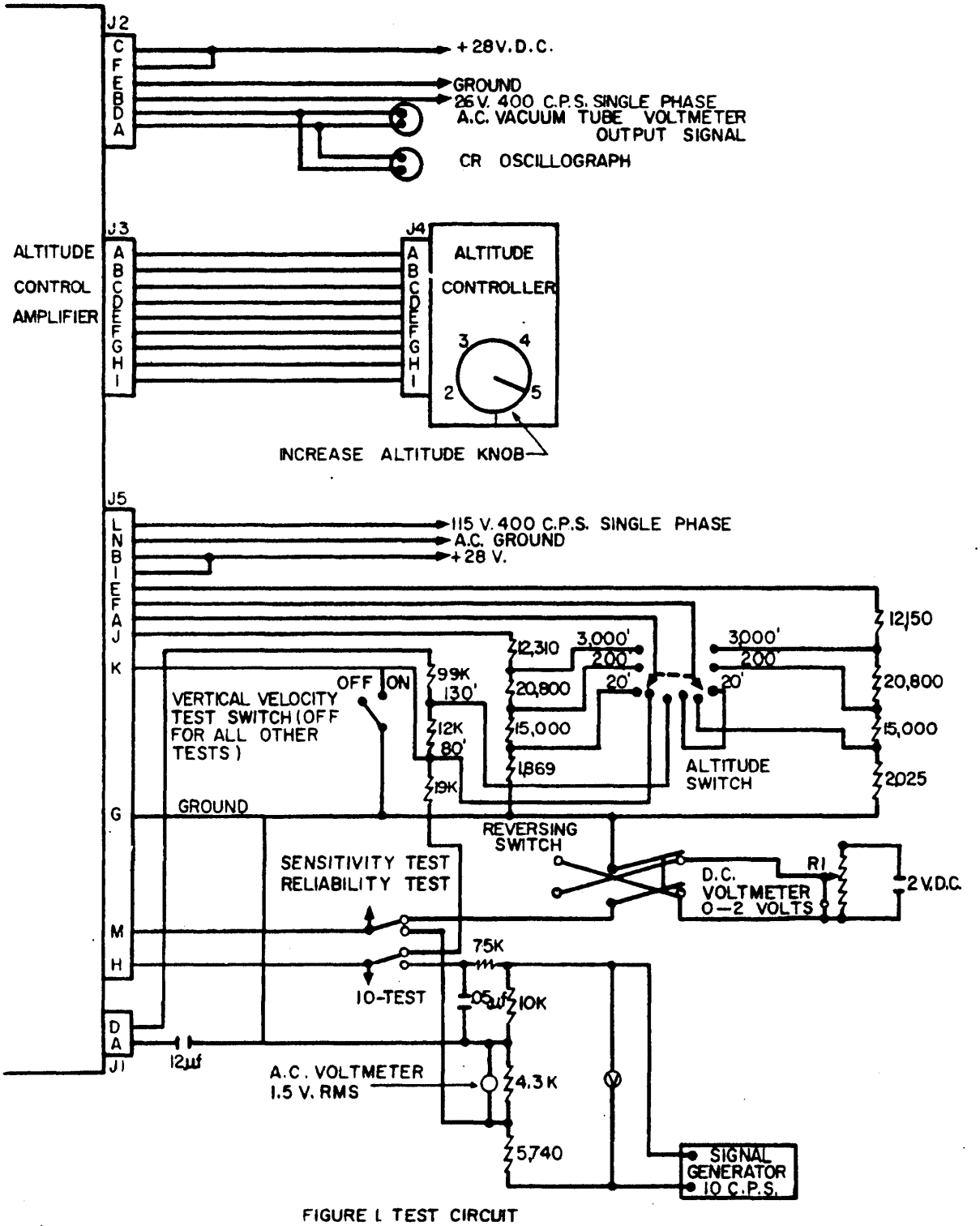


FIGURE 1. TEST CIRCUIT

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with altitude settings of 20 ft., 200 ft. and 3000 ft. selected at the altitude switch, allowing a stabilizing time of 2 minutes after each switch setting.

4.6.4 Circuit reliability

4.6.4.1 Altitude error - With the amplifier connected as specified in the Output signal test, but with zero input signal and the altitude switch in the 20 ft. position, the controller warning light shall be off and remain off when the altitude switch is changed from 20 ft. to 80 ft. The warning light shall light when the altitude switch is changed to 130 ft.

4.6.4.2 Vertical velocity - With the altitude switch at 20 ft. and terminals G and K connected together the signal light shall light when the velocity test switch is opened.

4.6.5 10 cycle test - The amplifier shall be connected as shown in Figure 1, with a 10 cycle signal between terminals M and ground and H and ground. With a signal of 1.5 volts rms between the M and ground the output signal between terminals A and D of J-2 shall not have a 10 cycle modulation of more than 10 millivolts rms.

4.6.6 Dielectric strength - The ground connections of the control amplifier shall be removed and a potential of 500 volts (root mean square) at commercial frequency shall be applied between each pin and ground for a period of 5 seconds. There shall be no breakdown of insulation or other permanent damage to the instrument as a result of this test.

4.6.7 Vibration error - The amplifier and mounting base shall be mounted on the vibration stand in its normal operating position and connected as shown in Figure 1. The amplifier shall be subjected to a vibration as specified herein with a circle diameter of 0.018 to 0.020 of an inch at frequencies from 300 to 3000 cycles per minute. In this frequency range the signal output as measured in accordance with the Output signal sensitivity test shall not vary by more than 10 percent of the value obtained under no vibration. There shall be no vibration frequency of the control amplifiers observed at any point in the frequency range.

4.6.8 Low temperature operation - (This test may be combined with the Low temperature exposure test at the discretion of the testing agency). The amplifier shall be properly connected except that no power shall be applied. The amplifier shall then be subjected to an ambient temperature of -54°C (-65°F) for a period of 4 hours. At the end of this period, and with the temperature maintained at -54°C (-65°F), the amplifier shall be energized and subjected to and meet the requirements specified in the Output signal test except that additional bias of plus 10 feet shall be added to the positive tolerance limit.

4.6.9 High temperature operation - The amplifier shall be properly connected except that no power shall be applied. The amplifier shall then be subjected to an ambient temperature of 71°C (160°F) for a period of 4 hours. At the end of this period, and with the temperature maintained at 71°C (160°F), the amplifier shall be energized for one minute and then subjected to and meet the requirements specified in the Output signal test.

4.6.10 Radio interference - The control amplifier and altitude controller shall be connected as shown in Figure 1 and subjected to and meet the requirements for conducted and radiated radio noise interference in accordance with Specification MIL-I-6181.

4.6.11 Low temperature exposure - The controller shall be tested in accordance with Procedure II of Specification MIL-E-5272. At the end of the 24 hour period, and while still at the low temperature, the controller shall operate without indications of erratic action. After the controller has saturated at room temperature for 4 hours, the controller shall be subjected to and meet the requirements specified in the following tests:

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(a) Continuity
(b) Output signal (sensitivity test)

4.6.12 High altitude - The altitude control amplifier shall be connected to the controller as shown in Figure 1. The amplifier shall be subjected to a pressure equivalent to 35,000 ± 500 feet at an ambient air temperature of -10° to -20°C (14° to -4°F) for a period of 3 hours. There shall be no damage to the amplifier which would affect subsequent operation.

4.6.13 Operation endurance - The control amplifier shall be connected to the altitude controller as shown in Figure 1, and the engaging switch closed. A one volt direct current shall be applied between pins M and G of receptacle J-5 to simulate the error signal obtained from the radar set for deviation from preset altitude. The system shall be operated continuously for 300 hours while the voltage across pins M and G of receptacle J-5 is reversed at the rate of two times per minute centering the servo for 30 seconds out of every 5 minutes. At the end of this period, the amplifier shall be subjected to and meet the requirements specified in the following tests:

(a) Continuity
(b) Output signal (sensitivity test)

4.6.14 Vibration endurance - The control amplifier and mounting base shall be mounted on a vibration stand and shall be subjected to vibration at a frequency of 2000 cycles per minute for a period of 100 hours so that a point on the stand will describe a circle 0.018 to 0.020 inch in diameter. The amplifier shall be energized at the end of this period. There shall be no evidence of damage as a result of the vibration, and the control amplifier shall be subjected to and meet the requirements specified in the following tests:

(a) Continuity
(b) Output signal (sensitivity test)

4.6.15 Humidity - The amplifier shall be tested in accordance with Procedure I of Specification MIL-E-5272 except that the test period shall be 120 hours (5 cycles). At the conclusion of the test, the amplifier shall be subjected to and meet the requirements specified in the Output signal test.

4.6.16 Fungus resistance - The amplifier shall be tested in accordance with Procedure I of Specification MIL-E-5272 except that the test period shall be 14 days. At the option of the qualifying agency, tests of component parts of the item may be accepted in lieu of, or in addition to, tests of the assembled instrument.

4.6.17 Salt spray - The amplifier shall be tested in accordance with Procedure I of Specification MIL E-5272. At the conclusion of the test, the amplifier shall be subjected to and meet the requirements specified in the following tests:

(a) Continuity test
(b) Output signal test

4.6.18 Flight test - The flight test shall consist of flying an airplane on the P-1 automatic pilot while interconnected to the radar set via the altitude control amplifier and altitude controller. The operation shall be within the limits specified in 3.3 of this specification.

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5. PREPARATION FOR DELIVERY

5.1 Application - The requirements of section 5 apply only to direct purchases or direct shipment to the Government.

5.2 Packaging

5.2.1 Level A - The amplifiers shall be packaged in accordance with MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract. For level A the method of preservation shall be MIL-P-116, Method IId using metal reusable containers.

5.2.2 Level C - When this level is required, packaging shall be in accordance with standard commercial practices.

5.3 Packing

5.3.1 Levels A and B - The amplifiers shall be packed in accordance with MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract.

5.3.2 Level C - The amplifiers shall be packed to ensure that the shipment arrives in satisfactory condition. The shipment shall conform to the applicable carrier rules and regulations in effect at time of shipment.

5.4 Marking - The interior and exterior containers shall be marked as specified in MIL-P-7936.

6. NOTES

6.1 Intended use - The altitude control amplifier covered by this specification is intended for use with the P-1 automatic pilot when automatic altitude control is to be maintained by means of the interconnection to the radar set AN/APN-22.

6.2 Ordering data - Requsitions, contracts and orders shall specify the NAF part number of the amplifier desired, the quantity, and whether overseas packing is to be furnished.

6.3 Preproduction - Manufacture of any further amplifiers on particular contract shall be suspended until the samples submitted are pronounced satisfactory by the Bureau of Aeronautics. When a contractor is in continuous production of these amplifiers from contract to contract, submission of further preproduction samples on the subsequent contract may be waived at the discretion of the Bureau of Aeronautics. Approval of preproduction samples for the waiving of preproduction tests does not preclude the requirements of acceptance testing.

6.3.1 It is understood that the products supplied under contract or order shall be identical to the corresponding preproduction samples in design, construction, quality, materials, workmanship and methods of manufacture. Deviation from the standards of the preproduction sample shall be made only by the Bureau of Aeronautics. Evidence of unauthorized change shall constitute cause for rejection.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-9279

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